Statistics for Data Science

Appendix to Group Project Report – Students' Academic Performance

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Fig. S1. Variables frequency distribution.

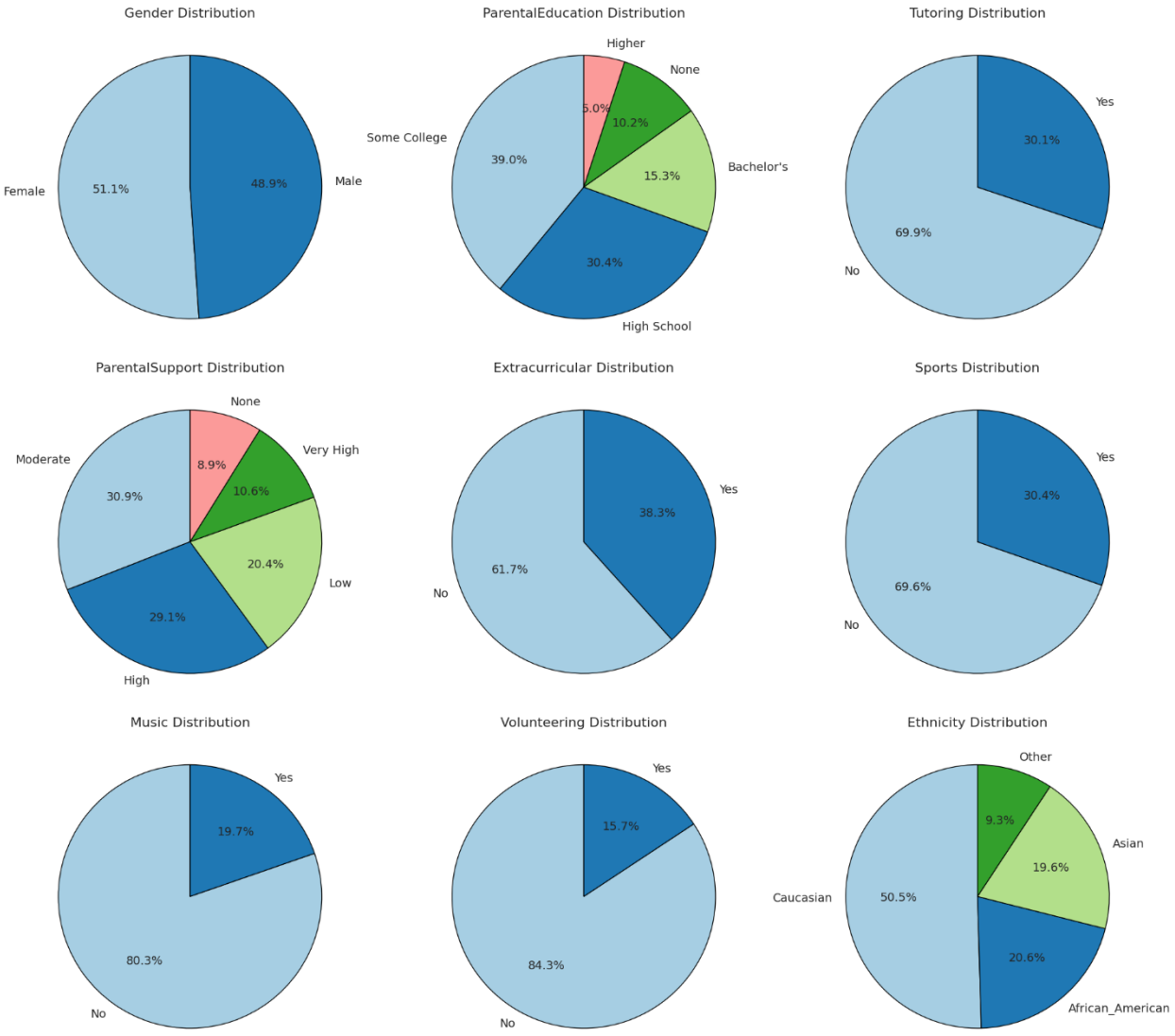


Fig. S2. Categorical Variables pie chart distribution.

Table S1. ANOVA for categorical variables.

	SUM_SQ	DF	F	PR(>F)
GENDER	0.357422	1	0.426665	0.513693
RESIDUAL	2002.129	2390	NaN	NaN
PARENTALEDUCATION	6.049132	4	1.80813	0.124484
RESIDUAL	1996.438	2387	NaN	NaN
TUTORING	42.17137	1	51.41498	9.92E-13
RESIDUAL	1960.315	2390	NaN	NaN
PARENTALSUPPORT	73.45501	4	22.72346	1.91E-18
RESIDUAL	1929.032	2387	NaN	NaN
EXTRACURRICULAR	17.72347	1	21.34214	0.000004
RESIDUAL	1984.763	2390	NaN	NaN
SPORTS	6.703673	1	8.027815	0.004645
RESIDUAL	1995.783	2390	NaN	NaN
MUSIC	10.76442	1	12.91695	0.000332
RESIDUAL	1991.722	2390	NaN	NaN
VOLUNTEERING	0.021254	1	0.025368	0.873468
RESIDUAL	2002.466	2390	NaN	NaN
ETHNICITY_CAUCASIAN	2.234142	1	2.669463	0.102422
RESIDUAL	2000.253	2390	NaN	NaN
ETHNICITY_AFRICAN_AMERICAN	1.006082	1	1.201379	0.273157
RESIDUAL	2001.481	2390	NaN	NaN
ETHNICITY_ASIAN	0.1486	1	0.17737	0.673681
RESIDUAL	2002.338	2390	NaN	NaN
ETHNICITY_OTHER	0.421646	1	0.503347	0.478102
RESIDUAL	2002.065	2390	NaN	NaN

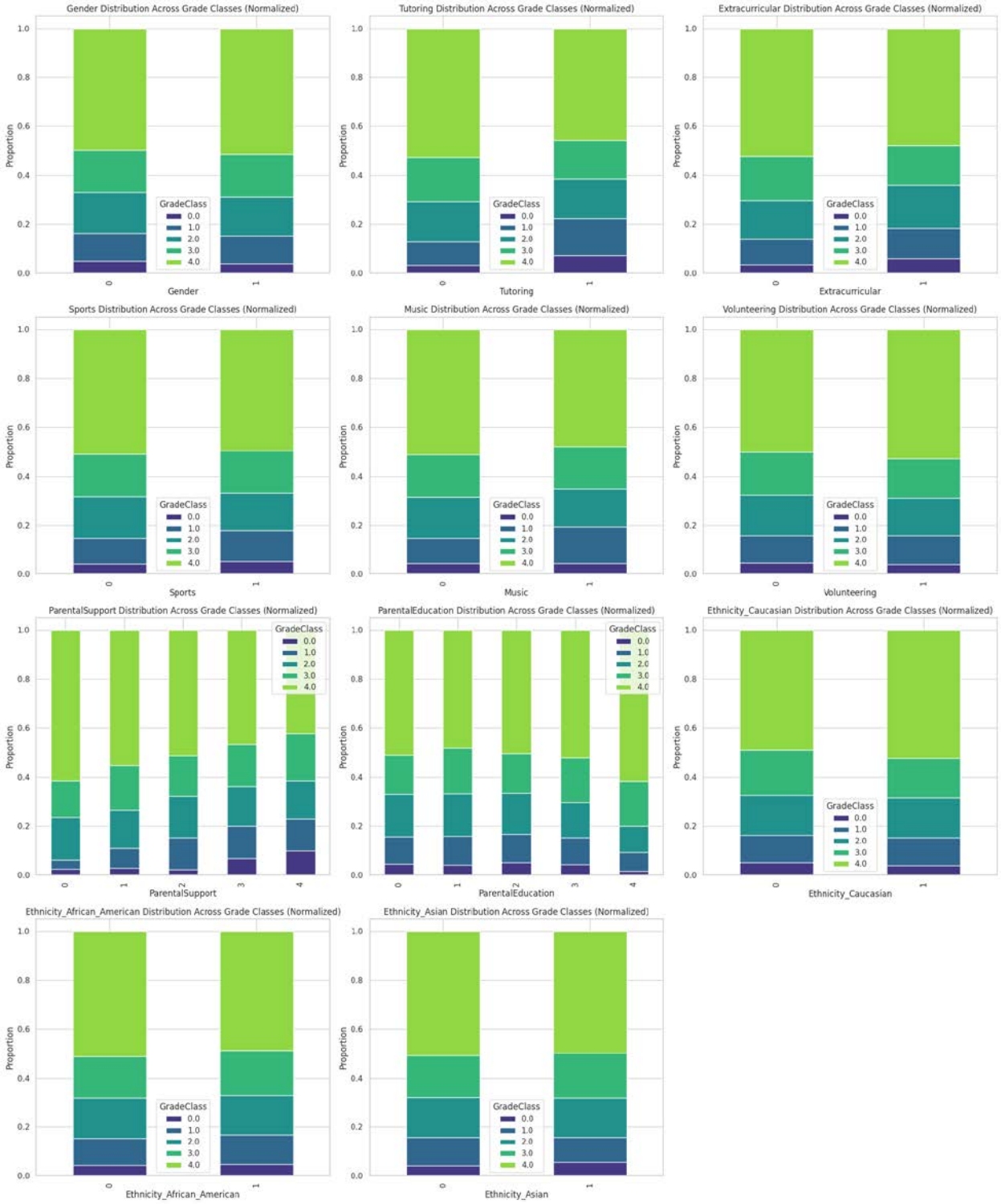


Fig. S3. *Normalized variable distribution across Grade Class levels.*